21.10.15.

Topic: First Order Computation in φ^4 Theory

$$S = \Omega_-^{-1}\Omega_+ = \lim_{\substack{t_1\to-\infty \\ t_2\to\infty}} U_0(-t_2)\,U(t_2-t_1)\,U_0(t_1)$$

$$H = \underset{\sim}{H_0} + gH_I$$

interaction

hamiltonian

Dyson Series

$$S = 1 + \sum_{n=1}^{\infty} \frac{(-ig)^n}{n!} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} T\, H(\theta_1) H(\theta_2) \cdots d\theta_n \cdots d\theta_1$$

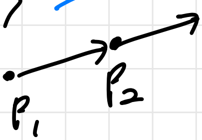
$H_I = V$, $\quad H_0 = -\frac{1}{2M}\Delta$ $\qquad H = \int \mathcal{H}$

$$S = 1 + (-ig)\int_{-\infty}^{\infty} e^{itH_0}\, V\, e^{-itH_0}\, dt + O(g^2)$$

Hamiltonian density

$$S = 1 + \sum_{n=0}^{\infty} \frac{(-ig)^n}{n!} \int_{\mathbb{R}^{1,3}} \int_{\mathbb{R}^{1,3}} \cdots T\, \mathcal{H}(x_1)\mathcal{H}(x_2) \cdots dx_n \cdots dx_1$$

$\langle p_2 | S | p_1 \rangle$ → probability of p_1 becomes p_2.



$$\langle \vec{p}_1, \vec{p}_2, \cdots, \vec{p}_k | S | \vec{q}_1, \cdots, \vec{q}_j \rangle$$

$$\langle \vec{p}_3, \vec{p}_4 | S | \vec{p}_2, \vec{p}_1 \rangle$$

Notation $\quad p = (\omega_p, \vec{p}) \in \mathbb{R}^{1,3}$ spacetime.

$$\omega_p = \sqrt{m^2 + |\vec{p}|^2}$$

$$|\vec{p}_1\rangle = \widehat{e^{i p_1 \cdot x}}$$

$$\langle \vec{p}_2 | \vec{p}_1 \rangle = (2\pi)^3 \delta^{(3)}(\vec{p}_2 - \vec{p}_1)$$

$$\mathcal{H}_{sym}^{\otimes 0} = \mathbb{C} \ni 1$$

Def $$|\vec{p}_j, \cdots, \vec{p}_1\rangle := a^\dagger(\vec{p}_j)\cdots a^\dagger(\vec{p}_1)\underbrace{|0\rangle}_{\in\, \mathcal{H}_{sym}^{\otimes n}}$$

$$\langle \vec{q}_1, \cdots, \vec{q}_j | := \langle 0 | a(\vec{q}_1)\cdots a(\vec{q}_j)$$

- $$\langle \vec{p}_3, \vec{p}_4 | \vec{p}_2\ \vec{p}_1 \rangle = \langle 0 | a(\vec{p}_3) a(\vec{p}_4) a^\dagger(\vec{p}_2) a^\dagger(\vec{p}_1) | 0 \rangle$$

φ_4 Theory

$$\mathcal{H} = \frac{1}{4!} :\varphi(x)^4:$$

$$\langle \vec{p}_3, \vec{p}_4 :\varphi^4: \vec{p}_2, \vec{p}_1 \rangle$$

$$= \langle 0 | a(\vec{p}_3) a(\vec{p}_4) :\varphi^4: a(\vec{p}_2) a(\vec{p}_1) | 0 \rangle$$

normal ordering term

I. Wick Theorem

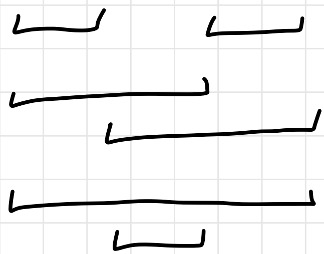
$$\mathcal{C} := \{A(\xi) + A^{\dagger}(\eta) : \xi, \eta \in \mathcal{H} = L^2(X_m, d\lambda_m)\}$$

<u>Def</u> A pairing l of $\overset{\text{even}}{k}$ to is a set of the form

$$\begin{cases} \{(l_1, l_1'), \cdots, (l_{k/2}, l_{k/2}')\} \text{ s.t. } l_j < l_j', \\ \{l_1, \cdots, l_k\} = \{1, 2, \cdots, k\} \end{cases}$$

ex) $k=4$

1 2 3 4



$\{(1\ 2), (3\ 4)\}$

$\{(1, 3), (2, 4)\}$

$\{(1, 4), (2, 3)\}$

<u>Rmk</u> # of parings of $k = (k-1)!!$

$= (k-1)(k-3)\cdots 5 \cdots 3 \cdot 1.$

<pf> $\dfrac{\binom{k}{2}\binom{k-2}{2}\cdots\binom{2}{2}}{\underbrace{(\frac{k}{2})!}_{\text{순서무시}}}$ $\square$

$a a a^{+} a^{+}$

$a(x) = A(\delta_x)$

Thm (Wick) $B_i \in C_c$, $i = 1, \cdots, k$.

$$\langle 0 | B_1 B_2 \cdots B_k | 0 \rangle = \begin{cases} \sum_{\text{pairings of } k} \prod_{i=1}^{k/2} \langle 0 | B_{l_i} B_{l_i'} | 0 \rangle & \text{even} \\ 0 & \text{if } k: \text{odd} \end{cases}$$

$$\langle 0 | B_1 B_2 B_3 B_4 | 0 \rangle = \langle 0 | B_1 B_2 | 0 \rangle \langle 0 | B_3 B_4 | 0 \rangle + \langle 0 | B_1 B_3 | 0 \rangle \langle 0 | B_2 B_4 | 0 \rangle + \langle 0 | B_1 B_4 | 0 \rangle \langle 0 | B_2 B_3 | 0 \rangle.$$

두개의 곱을 계산.

Lemma $B_1 := A(\xi_1) + A^{+}(\eta_1)$

$B_2 := A(\xi_2) + A^{+}(\eta_2)$

$\xi_i, \eta_i \in \mathcal{H} = L^2(X_m, d\mu_m)$

$$\langle 0 | B_1 B_2 | 0 \rangle = (\xi_1, \eta_2)$$

<pf> • Observe $\begin{cases} A(\xi) | 0 \rangle = 0 & \text{anhiliating empty sp} \\ \langle 0 | A^{+}(\eta) = 0. \end{cases}$

$$A(\xi) = \sum \alpha_k^* a_k$$

• $[A(\xi), A^{+}(\eta)] = (\xi, \eta) 1$

$$\langle 0 | B_1 B_2 | 0 \rangle = \langle 0 | A(\xi_1) A^{+}(\eta_2) + A(\xi_1) A(\xi_2) + A^{+}(\eta_1) A(\xi_2) + A^{+}(\eta_1) A(\eta_2) | 0 \rangle$$

$$= \langle 0 | A(\xi_1) A^{+}(\eta_2) | 0 \rangle$$

$AA^{+} = A^{+}A + [A, A^{+}]$

$$= \langle 0 | A^{+}(\eta_2) A(\xi_1) | 0 \rangle + \langle 0 | [A(\xi_1) A^{+}(\eta_2)] | 0 \rangle = (\xi_1, \eta_2).$$

$= 0$

Identity.

$A|0\rangle = 0$

$\langle 0|A^\dagger = 0$

< Proof of Wick's Thm >

<u>Induction on k.</u> $B_i = A(\xi_i) + A^\dagger(\eta_i)$

$$\langle 0|B_1B_2\cdots B_k|0\rangle = \langle 0|B_1\cdots B_{k-1}(A(\xi_k)+A^\dagger(\eta_k))|0\rangle$$

$$= \langle 0|B_1\cdots B_{k-2}(A(\xi_{k-1})+A^\dagger(\eta_{k-1}))A^\dagger(\eta_k)|0\rangle$$

$$= \langle 0|B_1\cdots B_{k-2}A^\dagger(\eta_k)A(\xi_{k-1})|0\rangle +$$

$[A(\xi), A^\dagger]$

$$= \langle 0|B_1\cdots B_{k-2}\boxed{(\xi_{k-1}, \eta_k)\mathbb{1}}|0\rangle$$

$[A^\dagger, A^\dagger] = 0$

anihilating operator commutes.

$$+\underbrace{\langle 0|B_1\cdots B_{k-2}A^\dagger(\eta_{k-1})A^\dagger(\eta_k)|0\rangle}_{}. \ \square$$

$(A(\xi_{k-1})+A^\dagger(\eta_{k-1}))$ $= (\xi_{k-1}, \eta_k)\langle 0|B_1\cdots B_{k-2}|0\rangle$

$p = (\omega_p, \vec{p})$

$a(p)$

$a(\vec{p}) = \dfrac{a(p)}{\sqrt{2\omega_{\vec{p}}}}$

<u>Lem</u> $\langle 0|a(\vec{p})\varphi(x)|0\rangle = \dfrac{1}{\sqrt{2\omega_{\vec{p}}}}e^{i(x,p)}$

$\langle 0|\varphi(x)a^\dagger(\vec{p})|0\rangle = \dfrac{1}{\sqrt{2\omega_{\vec{p}}}}e^{-i(x,p)}$

$\langle pf\rangle$ $\varphi(x) = \displaystyle\int_{\mathbb{R}^3}\frac{d\vec{p}}{(2\pi)^3 2\omega_{\vec{p}}}\left(e^{-i(x,p)}a(p) + e^{i(x,p)}a^\dagger(p)\right)$

$$= \int_{\mathbb{R}^3}\underbrace{\frac{d\vec{p}}{(2\pi)^3\sqrt{2\omega_{\vec{p}}}}}_{d\lambda_m}\left(e^{-i(x,p)}a(\vec{p}) + e^{i(x,p)}a^\dagger(\vec{p})\right)$$

$$= \int_{X_m} e^{-i(x,p)}a(\vec{p}) + e^{i(x,p)}a^\dagger(\vec{p})\, d\lambda_m$$

mass shell

$$= A(f_x) + A^\dagger(f_x)$$

where $f_x = e^{i(x,p)}$. $A(f) = \int f^* a$.

① $\langle 0 | a(\vec{p}) \varphi(x) | 0 \rangle$

$$= \langle 0 | \frac{A(\delta_p)}{\sqrt{2\omega_{\vec{p}}}} (A(f_x) + A^{\dagger}(f_x)) | 0 \rangle$$

$$= \frac{1}{\sqrt{2\omega_{\vec{p}}}} \langle 0 | A(\delta_p) A^{\dagger}(f_x) | 0 \rangle$$

$$= \frac{1}{\sqrt{2\omega_{\vec{p}}}} (\delta_p , f_x) = \frac{1}{\sqrt{2\omega_{\vec{p}}}} e^{i(x,p)} .$$

21. 10. 22.

Topic: Calcubtion of $\langle \vec{\mathbb{P}_3}, \vec{\mathbb{P}_4} | S | \vec{\mathbb{P}_1}, \vec{\mathbb{P}_2} \rangle$

$$S = 1 + (-ig)\int_{\mathbb{R}^{1,3}} \mathcal{H}(x)\,dx + \frac{(-ig)^2}{2!}\int_{\mathbb{R}^4}\int_{\mathbb{R}^4} \mathcal{T}\mathcal{H}(x_1)\mathcal{H}(x_2)\,dx_1 dx_2 + O(g^3)$$

$$\mathcal{H} = \frac{1}{4!} : \varphi(x)^4 :$$

$$\varphi(x) = \int_{\mathbb{R}^3} \frac{d\vec{p}}{(2\pi)^3\sqrt{2\omega_{\vec{p}}}} \left(e^{-i(x,p)} a(\vec{p}) + e^{i(x,p)} a^{+}(\vec{p}) \right)$$

p_1, p_2, p_3, p_4 : distinct.

$$\langle \vec{\mathbb{P}_3}, \vec{\mathbb{P}_4} | S | \vec{\mathbb{P}_1}, \vec{\mathbb{P}_2} \rangle$$

$$= \langle \vec{\mathbb{P}_3}, \vec{\mathbb{P}_4} | \vec{\mathbb{P}_1}, \vec{\mathbb{P}_2} \rangle + \frac{(-ig)}{4!}\int_{\mathbb{R}^{1,3}} \langle 0 | a(\vec{\mathbb{P}_3}) a(\vec{\mathbb{P}_4}) : \varphi(x)^4 : a^{+}(\vec{\mathbb{P}_1}) a^{+}(\vec{\mathbb{P}_2}) | 0 \rangle\, dx$$

$$+ \frac{(-ig)^2}{2!4!4!}\int_{\mathbb{R}^{1,3}}\int_{\mathbb{R}^{1,3}} \langle 0 | a(\vec{\mathbb{P}_3}) a(\vec{\mathbb{P}_4}) \mathcal{T} : \varphi(x_1)^4 : : \varphi(x_2)^4 : a^{+}(\vec{\mathbb{P}_2}) a^{+}(\vec{\mathbb{P}_1}) | 0 \rangle\, dx_1 dx_2 + O(g^3)$$

(Wick's Thm)

$$B_i = A(\xi_i) + A^{+}(\eta_i)$$

$$\langle 0 | B_1 B_2 \cdots B_k | 0 \rangle = \begin{cases} \sum_{\text{pairings}} \prod \langle 0 | B_{l_i} B_{l_i'} | 0 \rangle \\ \quad k : \text{odd}. \end{cases}$$

① $\langle 0| a(\vec{p}_3) a(\vec{p}_4) a^\dagger(\vec{p}_1) a^\dagger(\vec{p}_2)|0\rangle = 0$

$= A(\delta_{p_4}) \cdot \frac{1}{\sqrt{2\omega_{p_4}}}$

$\langle 0| a^\dagger = 0 = a|0\rangle$

$\langle 0| a(p) a^\dagger(q) |0\rangle = \langle 0| A(\delta_p) A^\dagger(\delta_q) |0\rangle = \delta(p-q)$

$\therefore$ If p_1, p_2, p_3, p_4 are distinct,

$\langle \vec{p}_3, \vec{p}_4 | \vec{p}_1, \vec{p}_2 \rangle = 0.$

I. Computation Involving Normal Ordering.

<u>Thm</u> $\langle 0| B_1 \cdots :B_i^k: \cdots B_m |0\rangle$

$= \sum_{\text{pairings}} \prod \langle 0| B_l B_k |0\rangle$

without ones which has $\langle 0| B_i B_i |0\rangle$.

ex) $\langle 0| B_1 :B_2^2: B_3 |0\rangle$

$\langle 0| B_1 \underbrace{:B_2^2:}_{\substack{\parallel \\ B_2 B_2}} B_3 |0\rangle = \langle 0| B_1 B_2 |0\rangle \langle 0| B_2 B_3 |0\rangle$

$+ \langle 0| B_1 B_2 |0\rangle \langle 0| B_2 B_3 |0\rangle$

$\cancel{+ \langle 0| B_1 B_3 |0\rangle \langle 0| B_2 B_2 |0\rangle}$

<Sketch of Pf>

$B_i = A(\xi_i) + A^\dagger(\eta_i)$, $:B_i^k: = \sum_{j=0}^{k} \binom{k}{j} (A^\dagger)^j A^{k-j}$

$\langle 0| B_1 \cdots :B_i^k: \cdots B_m |0\rangle$

$= \sum_{j=0}^{k} \binom{k}{j} \langle 0| B_1 \cdots (A^\dagger)^j A^{k-j} \cdots B_m |0\rangle$

Use $\underbrace{\langle B_1 B_i \rangle}_{\substack{\parallel \\ \langle B A_i^\dagger \rangle}} \langle B_2 B_i \rangle \underbrace{\langle B_i B_3 \rangle}_{\substack{\parallel \\ \langle A_i B_3 \rangle}}$

$$\langle \vec{p}_3, \vec{p}_4 \,|\, :\varphi(x)^4: \,|\, \vec{p}_1, \vec{p}_2 \rangle$$

$$= \langle 0 | a(\vec{p}_3) a(\vec{p}_4) :\varphi(x)^4: a^\dagger(\vec{p}_1) a^\dagger(\vec{p}_2) | 0 \rangle$$

$$\left(\varphi(x) = A(f_x) + A^\dagger(f_x), \quad f_x(p) = e^{i(x, p)} \right)$$ Minkowski Inner Product

$$= 4! \langle 0 | a(\vec{p}_3) \varphi(x) | 0 \rangle \langle a(\vec{p}_4) \varphi(x) \rangle \langle \varphi(x) a(\vec{p}_1) \rangle \langle \varphi(x) a(\vec{p}_2) \rangle$$

Use $\langle 0 | a(\vec{p}) \varphi(x) | 0 \rangle = \dfrac{1}{\sqrt{2\omega_{\vec{p}}}} e^{i(x,p)}$

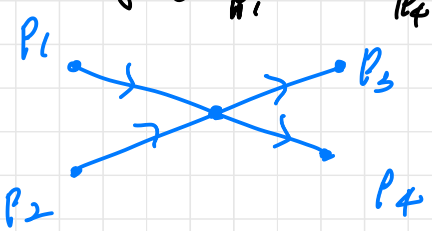
$\langle 0 | \varphi(x) a^\dagger(\vec{p}) | 0 \rangle = \dfrac{1}{\sqrt{2\omega_{\vec{p}}}} e^{-i(x,p)}$

$$= 4! \frac{e^{i(x, p_3 + p_4 - p_1 - p_2)}}{\sqrt{16 \omega_{\vec{p}_1} \omega_{\vec{p}_2} \omega_{\vec{p}_3} \omega_{\vec{p}_4}}}$$

$$\frac{-ig}{4!} \int_{\mathbb{R}^{1,3}} \frac{4!\, e^{i(x, \cdots)}}{\sqrt{16 \cdots}} dx$$

$p_3 + p_4 = p_1 + p_2$

$$= \frac{(-ig)\, \delta(p_3 + p_4 - p_1 - p_2)}{\sqrt{16 \omega_{\vec{p}_1} \cdots \omega_{\vec{p}_4}}} \cdot (2\pi)^4$$



$$\langle \vec{p}_3, \vec{p}_4 | S | \vec{p}_1, \vec{p}_2 \rangle$$

$$= \langle \vec{p}_3, \vec{p}_4 | \vec{p}_1, \vec{p}_2 \rangle + \frac{(-ig)}{4!} \int_{\mathbb{R}^{1,3}} \langle \vec{p}_3, \vec{p}_4 : \varphi(x)^4 : \vec{p}_1, \vec{p}_2 \rangle \, dx$$

$$+ \frac{(-ig)^2}{2!\,4!\,4!} \int_{\mathbb{R}^{1,3}} \int_{\mathbb{R}^{1,3}} \langle \overset{a}{\vec{p}_3}, \overset{a}{\vec{p}_4}\, T : \varphi(x_1)^4 : : \varphi(x_2)^4 : \overset{a^\dagger}{\vec{p}_1}\, \overset{a^\dagger}{\vec{p}_2} \rangle \, dx_1 dx_2$$

$$+ O(g^3) \qquad = \Sigma(\text{Type I}) + \Sigma(\text{Type II})$$

$$T A(t) A(s) = \begin{cases} A(t) A(s), & t \geq s \\ A(s) A(t), & t \leq s. \end{cases}$$

$$T \mathcal{H}(\underbrace{t, \vec{x}}_{x}) \mathcal{H}(\underbrace{s, \vec{y}}_{y}) = \begin{cases} \mathcal{H}(x) \mathcal{H}(y), & t \geq s \\ \mathcal{H}(y) \mathcal{H}(x), & t \leq s \end{cases}$$

<u>Type I</u>

$$\langle a \varphi \rangle \langle a \varphi \rangle \langle \tilde{\varphi} a^\dagger \rangle \langle \tilde{\varphi} a^\dagger \rangle \langle T \varphi \tilde{\varphi} \rangle^2$$

<u>Type II</u>

$$\langle a \varphi \rangle \langle a^\dagger \varphi \rangle \langle \tilde{\varphi} a \rangle \langle \tilde{\varphi} a^\dagger \rangle \langle T \varphi \tilde{\varphi} \rangle^2$$

<u># of Type I</u> Choice of φ & $\tilde{\varphi}$: 2

$\varphi_1, \varphi_2, \varphi_3, \varphi_4 \leftarrow$ Choice of $\langle a \varphi \rangle$: 4×3

$a \quad a$ — Choice of $\langle \tilde{\varphi} a^\dagger \rangle$: 4×3

4×3 (순서 생각) — Choice of $\langle \varphi \tilde{\varphi} \rangle$: 2

$$2 \times (4 \times 3) \times (4 \times 3) \times 2$$

As a prototype, we compute (type I)

$$\langle 0|\varphi(x_1)a^\dagger(\vec{p}_1)|0\rangle\langle 0|\varphi(x_1)a^\dagger(\vec{p}_3)|0\rangle$$

$$\cdot\langle 0|a(\vec{p}_3)\varphi(x_2)|0\rangle\langle 0|a(\vec{p}_4)\varphi(x_2)|0\rangle$$

$$\cdot\boxed{\langle 0|T\varphi(x_1)\varphi(x_2)|0\rangle^2}$$

But we can $\langle 0|\varphi(x)a^\dagger(p)|0\rangle = \dfrac{e^{-i(x,p)}}{\sqrt{2\omega_{\vec{p}}}}$.

$$\langle 0|\mathcal{T}\varphi(x_1)\varphi(x_2)|0\rangle \ldots (*) \qquad x_1=(t_1,\vec{x}_1), \quad x_2=(t_2,\vec{x}_2).$$

If $t_1 \geq t_2$,

$(*) = \langle 0|\varphi(x_1)\varphi(x_2)|0\rangle$ survive

$$= \langle 0|(\underbrace{A(f_{x_1}) + A^\dagger(f_{x_1}))(A(f_{x_2}) + A^\dagger(f_{x_2})}_{})|0\rangle$$

$$= \langle f_{x_1}, f_{x_2}\rangle \text{ in } L^2(X_m, d\lambda_m)$$

$$\langle f_{x_1}, f_{x_2}\rangle = \int_{X_m} e^{-i(x_1,p)}e^{i(x_2,p)}d\lambda_m(p).$$

$$= \int_{\mathbb{R}^3}\frac{e^{-i(t_1-t_2)\omega_{\vec{p}}+i(\vec{x}_1-\vec{x}_2)\cdot\vec{p}}}{(2\pi)^3 2\omega_{\vec{p}}}d\vec{p}.$$

$$\therefore (*) = \int_{\mathbb{R}^3}\frac{e^{-i|t_1-t_2|\omega_{\vec{p}}+i(\vec{x}_1-\vec{x}_2)\cdot\vec{p}}}{(2\pi)^3 2\omega_{\vec{p}}}d\vec{p}.$$

$$=: -i\Delta_F(x_1-x_2),$$

where $\Delta_F(x)$ is Feynman propagator

$$\Delta_F(x) = i\int_{\mathbb{R}^3} \frac{1}{(2\pi)^3 2\omega_{\vec{p}}} e^{-i|t|\omega_{\vec{p}} + i\vec{x}\cdot\vec{p}}\, d\vec{p}$$

<u>But</u> $\Delta_F(0) = \int_{\mathbb{R}^3} \frac{d\vec{p}}{(2\pi)^3 2\omega_{\vec{p}}}$ $= \sqrt{\|\vec{p}\|^2 + m^2}$

$\Delta_F(x)$ is a tempered distribution.

$$\int \Delta_F(x) f(x)\, dx = \int \Delta_F(x) f(x)\, dt\, d\vec{x}.$$

as a tempered distribution

<u>Lemma</u> (Formally)

$$\Delta_F(x) = \lim_{\varepsilon \to 0+} \int_{\mathbb{R}^{1,3}} \frac{dp}{(2\pi)^4} \cdot \frac{e^{-i(x,p)}}{-\|p\|^2 + m^2 - i\varepsilon}$$

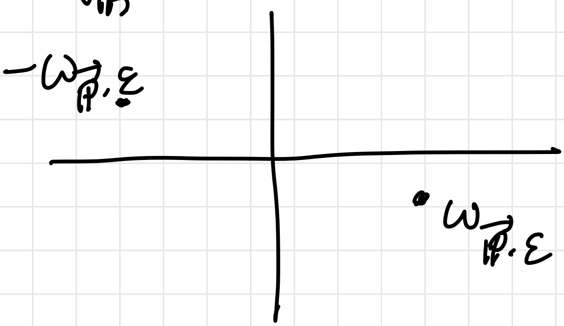
<typo>

<u>RHS</u> $\displaystyle \lim_{\varepsilon \to 0+} \int_{\mathbb{R}^3} \frac{e^{i(\vec{x}\cdot\vec{p})}}{(2\pi)^3} \left(\int_{\mathbb{R}} \frac{e^{-itp^0}\, dp^0}{(2\pi)(-(p^0)^2 + \|\vec{p}\|^2 + m^2 - i\varepsilon)} \right) d\vec{p}$

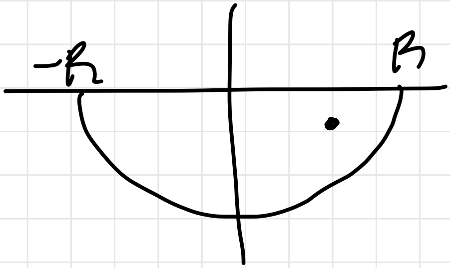
$\frac{1}{2\pi}\int_{\mathbb{R}} \frac{e^{-itz}}{f(z)}\, dz$, $\quad f(z) = -z^2 + \omega_{\vec{p}}^2 - i\varepsilon.$

Residue Thm

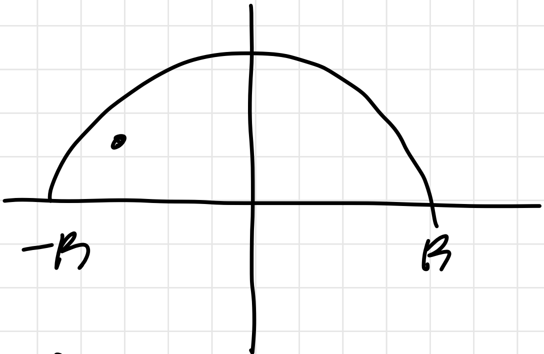
Jordan Lem.



$t \geq 0$



$t \leq 0$



$$\frac{1}{2\pi i}\int_{-\infty}^{\infty}\frac{e^{-itz}}{f(z)}\,dz = \frac{e^{-i|t|(\omega_{\vec{p},\varepsilon}}}{2\omega_{\vec{p},\varepsilon}}$$

inner product in space (not Minkowski)

$$\underline{\text{R.H.S}} = \lim_{\varepsilon\to 0}\int_{\mathbb{R}^3}\frac{e^{-i|t|(\omega_{\vec{p},\varepsilon}+i\vec{x}\cdot\vec{p}}}{(2\pi)^3\,2\omega_{\vec{p},\varepsilon}} = \Delta_F(x).$$

$$\langle 0|T\varphi(x_1)\varphi(x_2)|0\rangle^2 = \left(-i\Delta_F(x_1-x_2)\right)^2$$

$$= -\lim_{\varepsilon\to 0+}\int_{\mathbb{R}^{1,3}}\int_{\mathbb{R}^{1,3}}\frac{e^{-i(x_1-x_2,p)}\,e^{-i(x_1-x_2,q)}}{(2\pi)^8(-\underbrace{\|p\|}_{\text{Minkowski norm}}{}^2+m^2-i\varepsilon)(-\|q\|^2+m^2-i\varepsilon)}\,dp\,dq$$

$$\langle 0|\varphi(x_1)a^\dagger(\vec{p}_1)|0\rangle\langle 0|\varphi(x_1)a^\dagger(\vec{p}_2)|0\rangle$$
$$\cdot\langle 0|a(\vec{p}_3)\varphi(x_2)|0\rangle\langle 0|a(\vec{p}_4)\varphi(x_2)|0\rangle$$
$$\cdot\langle 0|T\varphi(x_1)\varphi(x_2)|0\rangle = (**).$$

$$(**) = \frac{-1}{\sqrt{16\omega_{\vec{p}_1}\cdots\omega_{\vec{p}_4}}}\lim_{\varepsilon\to 0+}\int_{\mathbb{R}^{1,3}}\int_{\mathbb{R}^{1,3}}\frac{e^{-i(x_1,\,p_1+p_2+p.q)}\,e^{i(x_2,\,p_3+p_4+p+q)}}{(2\pi)^8(-\|p\|^2+m^2-i\varepsilon)(-\|q\|^2+m^2-i\varepsilon)}\,dp\,dq$$

$$= \int_{\mathbb{R}^{1,3}}\int_{\mathbb{R}^{1,3}}\frac{\delta^{(4)}(p_1+p_2+p+q)\,\delta^{(4)}(p_3+p_4+p+q)}{(-\|p\|^2+m^2-i\varepsilon)(-\|q\|^2+m^2-i\varepsilon)}\,dp\,dq \quad \cdots (*)$$

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{ix\cdot\xi}\,d\xi = \delta(x) \quad (\text{Fourier})$$

$$(*) = \int\frac{\delta^{(4)}(p_3+p_4-p_1-p_2)\,dp}{(-\|p\|^2+m^2-i\varepsilon)(-\|-p_1-p_2-p\|^2+m^2-i\varepsilon)}$$

$$\int_{-\infty}^{\infty}\delta(x-z)\,\delta(y-z)\,\xi(z)\,dz = \delta(x-y)\,\xi(x)$$

In summary,

$$\langle \vec{p}_3, \vec{p}_4 | S | \vec{p}_1, \vec{p}_2 \rangle = \left(g - i g^2 M + O(g^3)\right) \frac{i(2\pi)^4 \delta^{(4)}(p_3 + p_4 - p_1 - p_2)}{\sqrt{16\,\omega_{\vec{p}_1} \cdots \omega_{\vec{p}_4}}}$$

where

$$M = C \sum \lim_{\varepsilon \to 0^+} \frac{1}{(2\pi)^4} \boxed{\int_{\mathbb{R}^4} \frac{\theta(\vec{p})\, dp}{(-\|p\|^2 + m^2 - i\varepsilon)(-\|p_1 + p_2 - p\|^2 + m^2 - i\varepsilon)}}$$

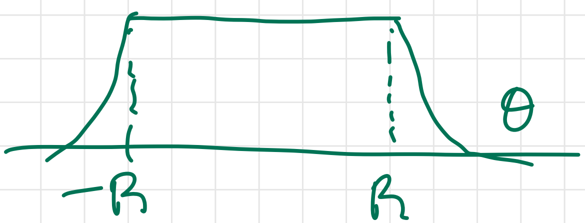
$p \longleftrightarrow -p$ change of variable

$$\sim \int_{\mathbb{R}^4} \frac{1}{\|p\|^4} = \infty$$

No way!

So we add

where $\theta(\vec{p}) = 1$ if $|\vec{p}| < R$ and decay fast outside of it.



Renormalization

Problem : We don't know θ.

sol : We don't have to know, provided that for special $(p_1^*, p_2^*, p_3^*, p_4^*)$ if we know $g - ig\tilde{\mathcal{M}}_* + \mathcal{O}(g^3)$.